

Andrés Pedraza Míguez

Senior Java & Platform Engineer | Agentic Orchestration & Distributed Systems

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|------------|--------------------------|---------|-----------------|
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PROFESSIONAL SUMMARY

Senior Backend & Platform Engineer with over 14 years of experience architecting resilient distributed systems, event-driven platforms, and high-throughput microservices using Java (Java 21, Spring Boot) and Kotlin. Deep practical expertise in Agentic Engineering, LLM orchestration, and Model Context Protocol (MCP) tool integration alongside production event streaming with Apache Kafka.

Proven track record leading zero-to-one modules and modernising mission-critical platforms in remote and UK enterprise environments (8+ years). Strong advocate of clean code, automated testing (TDD), observability, and API-First / Event-Driven contracts (OpenAPI, AsyncAPI). Skilled in driving technical proposals, designing fault-tolerant distributed workflows, and bridging deterministic backend infrastructure with autonomous AI agent capabilities.

WORK EXPERIENCE

Senior Java Developer (Acting Tech Lead) at Plexus Tech

Client: Inditex (Zara) | October 2023 – Present | Seville (Remote)

- Event-Driven Backend & Messaging:** Engineered high-resilience microservices using **Java 21** and **Spring Boot 3**, enforcing an API-First and Event-Driven architecture: designed **AsyncAPI** event contracts with **Apache Kafka & Avro** schemas for type-safe, asynchronous event streaming across global Zara warehouse logistics.
- Agentic AI & LLM Tooling Integration:** Architected and integrated agentic AI developer infrastructure (MCP - Model Context Protocol servers, OpenCode, GitHub Copilot) with custom tools, rules, and prompt engineering—automating technical system designs (SDD), PR reviews, test generation, and complex production telemetry investigations.
- Architectural Ownership & Clean Code:** Refactored complex distributed legacy modules into clean, decoupled Hexagonal / Clean Architecture patterns, drastically reducing technical debt and production incidents in a Dockerised Azure cloud ecosystem.
- Reliability & Testing:** Enforced rigorous testing strategies (JUnit 5, Mockito, contract testing, integration tests) and observability practices to guarantee deterministic behavior, fault tolerance, and zero regression across mission-critical logistics operations.
- Technical Leadership & Mentorship:** Mentored senior engineers on modern engineering practices, clean code, and API standards, facilitating technical alignment and RFC reviews across multi-disciplinary teams.

**Main Project: Raw Materials (Inditex)** A mission-critical event-driven platform managing real-time logistics, stock movements, and distribution of raw materials across global Zara supply chains.

TECH STACK

|         |               |              |                 |          |         |                              |        |        |       |
|---------|---------------|--------------|-----------------|----------|---------|------------------------------|--------|--------|-------|
| Java 21 | Spring Boot 3 | Apache Kafka | Avro            | AsyncAPI | OpenAPI | MCP (Model Context Protocol) | Kotlin | Docker | Azure |
| MongoDB | JUnit 5       | Mockito      | CI/CD pipelines |          |         |                              |        |        |       |

Senior Java Developer at Brandworkz

February 2017 – August 2023 | London (Hybrid)

- 0-to-1 Platform & API Architecture:** Designed and built core Java backend services from scratch within a high-throughput multi-tenant SaaS architecture serving enterprise clients (Boeing, Dr. Martens, Hitachi).
- Event & Data Streaming / Search:** Architected a high-performance reporting and data ingestion engine utilizing **Elasticsearch**, **Kibana**, and **Redis** for real-time indexing, querying, and analytics.

- **Resilience, Auth & Security:** Implemented robust **OAuth2** authentication/authorisation, token management, and role-based access control (RBAC), securing multi-tenant APIs against unauthorized access.
- **Cloud & DevOps (Build–Ship–Run):** Automated CI/CD pipelines, containerised services, and managed AWS infrastructure (EC2, S3, RDS) using Jenkins, Bash scripts, and monitoring tools to ensure 99.9% uptime.
- **System Design & Mentorship:** Authored technical specifications, led architectural trade-off discussions, and mentored junior and mid-level engineers in TDD, clean code, and cloud-native design.

**Main Project: Brandworkz SaaS Platform** A global Digital Asset Management (DAM) platform handling millions of digital assets, multi-tenant workflows, and distributed background processing for international enterprises.

#### TECH STACK

Java Spring Boot Spring Cloud Spring Security AWS (EC2/S3/RDS) Elasticsearch Redis PostgreSQL OpenAPI 3  
Liquibase Jenkins Docker

### Freelance Backend Developer (Contractor at Brandworkz)

February 2022 – August 2023 | Remote

- **Contracting & Autonomous Ownership:** Transitioned seamlessly to an autonomous contractor role, independently driving core backend performance optimizations, system refactoring, and database tuning during major multi-tenant architectural upgrades.
- **Latency & Performance Optimization:** Identified and resolved critical bottlenecks in distributed data pipelines, optimizing SQL queries, cache eviction policies, and async task execution.

### Back-end Developer at Locassa Ltd

July 2015 – November 2016 | London (On-site)

- **End-to-End Lifecycle & Delivery:** Managed full build-ship-run lifecycles for multiple client products and communication-enabled applications from initial technical design to production deployment and monitoring.
- **REST APIs & Third-Party Integrations:** Designed clean REST APIs and integrated third-party communication, notification, and payment gateways for startup backends.
- **Tech Leadership:** Mentored developers in modern Spring practices, automated testing, and agile delivery.

#### Projects:

Ziferblat, Minted, Inkvite, Jarvis, Vesta Fitness, Rounds on Me.

TECH STACK Spring Boot REST APIs MySQL OAuth2 AWS (Elastic Beanstalk S3 EC2)

### Java Analyst at Grupo Servinform

October 2014 – May 2015 | Seville (On-site)

- **Requirements Analysis:** Gathered requirements and authored technical analysis documents for government modernization projects.
- **Release & Deployment:** Led configuration, testing, and deployment in integration environments.

TECH STACK iBatis Spring Struts PL/SQL Apache Tomcat Maven

### Java Developer at Grupo Servinform

January 2012 – September 2014 | Seville (On-site)

- **Core Development:** Delivered new functionalities for the “HAPR” system (Telematic Authorisation) for the Junta de Andalucía.
- **Database Tuning:** Optimised database performance using advanced PL/SQL scripts.
- **Agile Delivery:** Applied Scrum methodology in an agile team environment.

TECH STACK Hibernate iBatis Spring JSP PL/SQL Subversion

## Junior Oracle/Java Developer at Ayesa

April 2011 – December 2011 | Seville (Part-time)

- **GUI & Legacy Maintenance:** Developed GUI prototypes and maintained legacy systems for public healthcare and social services projects.

### TECH STACK

Oracle Forms

PL/SQL

Hibernate

Spring

JSF

### SKILLS

- **Programming Languages:** Java (Expert - Java 21/17/8), Kotlin (Advanced), TypeScript/JavaScript (Proficient), SQL, Bash
- **Spoken Languages:** English (Full Professional / Fluent — 8+ years UK experience), Spanish (Native)
- **Core Frameworks & Backend:** Spring Boot 3, Spring AI, Spring Cloud, Hexagonal Architecture, Microservices, Clean Architecture, REST APIs
- **AI & Agentic Orchestration:** Agentic Workflows, Model Context Protocol (MCP), Custom Agent Skills & Rules, Tool & Function Calling, Prompt Engineering, Context Management, Guardrails & Error Recovery, Non-Deterministic Systems Testing, LLM Integration (Claude, GPT, Gemini)
- **Messaging & Event-Driven:** Apache Kafka, Avro, AsyncAPI, Event-Driven Architecture, Distributed Workflows
- **Data & Search:** PostgreSQL, MongoDB, Redis, Elasticsearch, Kibana, MySQL, Oracle, Liquibase
- **Cloud & DevOps:** AWS (EC2, S3, RDS), Azure, Docker, Kubernetes, CI/CD (Jenkins, GitHub Actions), GitOps, Observability & Monitoring
- **Testing & Quality:** JUnit 5, Mockito, TDD, Contract Testing, Code Reviews, API-First Design (OpenAPI Generator)

### EDUCATION

#### HNC in Administration and Information Technology

IES Punta del Verde | Graduated 2011

- Focused on software development and database management.
- Capstone project combining Java and SQL technologies.